SPACEBOX

Server & Dashboard Guide

Game version 0.1.5

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1. Start a server

The game, the host mode and the dedicated server are the same program. A dedicated server runs without a window and keeps the world running while nobody plays on that PC.

Windows

1. Open the game folder.
2. Double-click `start-server.bat`. A terminal window opens and shows the log.
3. Allow Spacebox in the Windows Firewall question (also on public networks if you want players from outside).

Linux

1. Unzip the Linux build.
2. Run `./start-server.sh` in a terminal.
3. No desktop, X11, Wayland or graphics card is needed: the server runs without a window.

On the first start the server creates `Multiplayer/serverconfig.json` and a new world in `Worlds/ServerWorld`. When the log shows `World ready`, players can join.

> **Stop the server safely**
> Type `stop` in the terminal (or use Stop in the dashboard). The world is saved first. Ctrl+C, closing the terminal window and a system stop signal (SIGTERM, for example from systemd or Docker) also save before the server stops (Windows waits only about 5 seconds when the window is closed). Killing the process loses changes since the last autosave.

Without the scripts: `Spacebox.exe --server` (Windows) or `./Spacebox --server` (Linux).

2. The config file

`serverconfig.json` lies in the `Multiplayer` folder next to the executable (together with this guide). Edit it while the server is stopped (most settings can also be changed live in the dashboard). Missing settings are added with their defaults on the next start.

Setting	Default	Meaning
ServerName	Spacebox Server	Name in the server list
Description	A Spacebox server	Text under the name (up to 600 characters)
IconFile	server-icon.png	PNG or JPEG in the `Multiplayer` folder, up to 256 KB, square 64–128 px looks best
LanDiscovery	true	Players in the same network find the server automatically
WhitelistEnabled / Whitelist	false / []	Only listed names may join
RegistrationEnabled	true	Players need a password (created on their first join)

Admins	[]	Player names with admin rights: they run every console command from the in-game chat with `/` (e.g. `/gm 1 Anna`, `/kick Bob`). Only works with `RegistrationEnabled`, so nobody can take an admin's name. The account of an admin name can only be created from the server's local network (or the server PC); after that the admin can log in from anywhere. Creative players and admins can also place shared ships (`.ship` files, Ships window on B); the server builds them as new ships
ShowAdmins	true	Show `[ADMIN]` before admin names in the player list and in chat
WorldName	ServerWorld	Folder in `Worlds/`; a missing world is created
Seed	random	World seed for a new world
ModId	default	GameSet (content pack) of the world
TexturePacks	[]	Texture pack ids from `Mods/TexturePacks/` sent to every player; they see them below their own packs while on this server
Port	7777	UDP port of the game (the dashboard uses TCP on the same number)
MaxPlayers	16	Player limit
GameMode	1	0 Survival, 1 Creative, 2 Spectator (for a new world)
AutoSaveInterval	300	Seconds between autosaves, 0 turns it off (minimum 10)
UpdateRate	60	Server frame rate while players are online, 10–240
IdleUpdateRate	10	Frame rate after 30 seconds without any connection, to save energy (0 = always use `UpdateRate`). Any connection switches back at once
SleepAfterMinutes	0	When nobody was online for this many minutes, the server saves and unloads the world but keeps the port open and stays in the server list. The first player who joins sees "The server is starting, please wait", the world loads in a few seconds and the game joins by itself. 0 = off

Backups	on	Automatic zip backups of the world in `Worlds/_backups/`: `Enabled` (true), `IntervalMinutes` (360, at least 10; skipped when nobody played since the last one), `KeepLast` (10 automatic and stop backups are kept, older ones are deleted; manual backups are never deleted), `OnStop` (true: one more backup when the server stops)
Claims	see chapter 7	Claim beacon settings: `Enabled`, `Radius` (asteroid zone ± blocks, 12), `MaxStoredMinutes` (43200 = 30 days), `WarningMinutes` (60), `ExtraClaimCostPercent` (50: every further claim of a player makes all of them use time 50 % faster), `MaxClaimsPerPlayer` (0 = unlimited), `MaxMembers` (16), `MinDistanceFromSpawn` (0 = off), `Payments` (ingot id → minutes: aluminium 30, iron 30, cobalt 60, titanium 600, thulium 720, uranium 1440, obscurium 10080). Also editable on the dashboard Settings page
Gameplay	all true	Gameplay rules: `Pvp` (players can hurt other players with weapons, drills, ship cannons and bombs), `Explosions` (bombs and power blocks explode at all), `ExplosionBlockDamage` (explosions break blocks), `ExplosionPlayerDamage` (explosions hurt players), `ExplosionsRespectClaims` (explosions do not break blocks in claims and protected areas), `KeepInventory` (true: players keep their items when they die; false: survival players drop hotbar, inventory and equipment where they die), `DeathDropMinutes` (10: how long those items stay before they disappear, 1–1440), `MaxShipSpeed` (0 = the GameSet's ship speed; any other value caps every ship at that many m/s, up to 1000). Also on the dashboard Settings page and with `gamerule`; changes apply at once
FriendlyMobs	false	Mobs never attack first; a mob that is hurt fights back for 30 seconds
DashboardEnabled	true	Web dashboard on or off
DashboardPort	0	TCP port of the dashboard, 0 = same number as `Port`
PublicAddress	empty	The address people use from the internet: a DuckDNS name, any domain or your public IP. Turns on automatic HTTPS (chapter 5)

DashboardAllowPublicHttp	false	Without HTTPS the dashboard only answers the local network. `true` also allows plain `http://` from the internet (passwords travel unencrypted, not recommended)
HttpsAutomatic	true	Get and renew the HTTPS certificate automatically when `PublicAddress` is set
HttpsChallengePort	80	Port for the Let's Encrypt check when no DNS token is set
DuckDnsToken	empty	For `*.duckdns.org` addresses: no port 80 needed, and the server keeps the name on your current IP
DuckDnsUpdateIp	true	Update the DuckDNS IP every 5 minutes
CloudflareApiToken	empty	For domains managed by Cloudflare: no port 80 needed
LetsEncryptEmail	empty	Optional, Let's Encrypt sends expiry warnings there
LetsEncryptStaging	false	Test certificates (not trusted by browsers), only for testing the setup
DashboardCertificateFile / ...KeyFile / ...Password	empty	Your own certificate files instead (PEM + key, or PFX + password); turns the automatic mode off

> **Cheat protection**
> The server counts the items of every survival player. Items only come from the server (picking up drops, taking from chests, crafting, gifts, `give`, tutorial rewards); a player whose game sends items the server never gave gets them removed again and the server log shows `[Items]`. Creative players are not checked.

3. Join the server

- **Same network:** Multiplayer menu → the server appears under local servers.
- **By address:** add a server with the IP (or DuckDNS name) and port 7777.
- **Same PC:** address `127.0.0.1`.

Players get the server's GameSet automatically if they do not have it. The game version must be the same on server and clients.

4. Open it to the internet

1. **Fixed LAN address:** give the server PC a fixed local IP in your router (DHCP reservation), e.g. `192.168.1.50`.
2. **Port forwarding:** in the router forward **UDP 7777** (game) and **TCP 7777** (dashboard) to that IP. Some routers have one rule for "TCP/UDP".

3. **Firewall:** Windows: allow Spacebox when asked, or add inbound rules for TCP and UDP 7777. Linux with ufw: `sudo ufw allow 7777`.
4. **A name instead of an IP:** home IPs change. A free DuckDNS name is easiest (chapter 5): the server keeps it up to date by itself. Any other name service works too.

> **CGNAT**
> If the WAN/Internet IP in your router differs from the IP that duckdns.org shows for you, your provider uses CGNAT and port forwarding cannot work. Ask the provider for a public IPv4 address, or use a VPN/tunnel service (for example a VPS or a mesh VPN for your friends).

> **Inside your own network**
> Many routers cannot open your own public name from inside (no "NAT loopback"). At home use the LAN address, e.g. `192.168.1.50:7777`; friends outside use the DuckDNS name.

5. Automatic HTTPS

The dashboard asks for your admin password. Over the internet it must be encrypted (HTTPS), otherwise the password can be read on the way. The server gets a free certificate from Let's Encrypt and renews it by itself. You only tell it the address people use, and pick one of the ways below.

Your situation	What to put in serverconfig.json	Extra port
Free DuckDNS name	`PublicAddress` + `DuckDnsToken`	none
Own domain on Cloudflare	`PublicAddress` + `CloudflareApiToken`	none
Any other name (No-IP, Dynu, own domain, ...)	`PublicAddress`	TCP 80
Only your public IP	`PublicAddress` = the IP	TCP 80

A. DuckDNS (easiest, free)

1. Open `https://www.duckdns.org`, sign in and add a sub domain, for example `myserver`.
2. Copy the **token** shown at the top of the page.
3. Stop the server and set:

```
"PublicAddress": "myserver.duckdns.org",
"DuckDnsToken": "a1b2c3d4-....-your-token"
```

4. Start the server. The server also keeps the name on your current IP, so the DuckDNS updater program is not needed.

B. Your own domain on Cloudflare

1. In Cloudflare, add a DNS record (A) like `play.example.com` → your public IP, proxy **off** (grey cloud).
2. My Profile → API Tokens → Create Token → template **Edit zone DNS**, zone = your domain. Copy the token.

3. Set:

```
"PublicAddress": "play.example.com",
"CloudflareApiToken": "your-cloudflare-token"
```

C. Any other name or just your IP

1. Make sure the name points to your public IP (No-IP, Dynu and others have their own small updater program).
2. Forward **TCP 80** to the server PC as well (Let's Encrypt checks a file there; the server opens port 80 only for those few seconds).
3. Set:

```
"PublicAddress": "play.example.com"     or your IP, e.g. "85.10.20.30"
```

Port 80 not possible?
Some providers block it, and on Linux only root may open ports below 1024. Forward external port 80 to another internal port in the router (for example 8080) and set `"HttpsChallengePort": 8080`. If your provider blocks port 80 completely, use DuckDNS or Cloudflare instead.

Only an IP address
Let's Encrypt gives certificates for IP addresses that last about 6 days; the server renews them automatically every few days. If your IP changes, a name (DuckDNS) is the better choice.

Then

Start the server. After about a minute the log shows `HTTPS certificate for … saved`, and the dashboard overview shows the HTTPS state. Open `https://<your address>:7777/dashboard` from anywhere.

What happens in the background
The certificate is stored in `Multiplayer/Dashboard/certificates/`, checked every 12 hours and renewed when a third of its time is left (30 days for normal certificates). Failures are written to the log with a hint and retried after an hour. Set `"LetsEncryptStaging": true` for a first test run: Let's Encrypt then issues test certificates (browsers warn about them) and failed attempts do not count against its limits.

Keep tokens secret
DuckDNS and Cloudflare tokens control your name. Do not share `serverconfig.json` or screenshots of it.

While HTTPS is active, plain `http://` still works inside your own network; from the internet it is redirected to `https://`. Without HTTPS the dashboard refuses plain `http://` from the internet unless `DashboardAllowPublicHttp` is true. The dashboard only answers addresses it knows (IP addresses, local names, `PublicAddress` and the certificate names); for another domain name set it as `PublicAddress`. Using your own certificate instead: set `DashboardCertificateFile` and `DashboardCertificateKeyFile`.

6. The web dashboard

First start: create the admin

1. Start the server. The log says: *No admin yet. Register in the dashboard with the setup key from …/Multiplayer/Dashboard/setup-key.txt.*
2. Open the dashboard: on the server PC `http://localhost:7777/dashboard`, in your network `http://192.168.1.50:7777/dashboard` (the server log prints the exact local addresses at start; Windows Firewall on the server PC must allow the server, TCP port 7777 for the dashboard), from outside `https://<your address>:7777/dashboard`.
3. Open `Multiplayer/Dashboard/setup-key.txt` in the server folder and copy the key (like `ABCDE-FGHJK-LMNPQ-RSTUV`).
4. Enter the key, an admin name and a password (at least 8 characters). The key file is deleted; only one admin exists.

> **Forgot the password?**
> Stop the server, delete `Multiplayer/Dashboard/admin.json` and start again: a new setup key is written and you can register again.

Pages

Page	What you can do
Overview	Players, uptime, CPU, memory, game ticks, update time, world size, live charts; server info and HTTPS state; send a chat message to everyone; Save world, Restart, Stop
Players	Online players with ping, address and position; all players who ever joined; Kick, Mute, Ban; change a player's game mode (also for offline players, kept when they rejoin, stored as `GameMode` in the player's file in `players/`). Click a player for health, power, oxygen, equipment, hotbar, inventory with item icons and statistics
Map	3D map of the loaded space around the players (500 to 4000 m, or around one player): asteroids as dots sized by their radius, ships as arrows, players with names. Colours show what is only regenerated from the seed, what players changed and saved, and unsaved changes; rings show active (gold) and expired (dashed) claims with the owner. Drag to rotate, right drag to pan, wheel to zoom, click a dot for details
Moderation	Banned names and IP addresses (unban), muted players, ban by name, whitelist on/off, add and remove names; admins: add and remove, show or hide the [ADMIN] tag
Console	Live server log; type any console command (Tab completes, arrow keys browse the history)
Settings	Name, description, max players, LAN visibility, autosave interval, frame rate (applied at once); gameplay rules (PvP, explosions, friendly mobs, keep inventory, death drop time); upload the server icon; change the dashboard password

> **Security built in**
> Login sessions last 12 hours after the last use and at most 7 days; 5 wrong passwords lock that address (IPv6: its /64 network) for 5 minutes, and at most 30 login attempts per minute are accepted from the internet; every admin action is written to the server log as `[Dashboard] admin: …`. If you never need the dashboard from outside, simply do not forward TCP 7777, or set `"DashboardEnabled": false`.

7. Console commands

Type them in the server terminal or the dashboard console. `help` lists all of them. Admins (see `Admins` in chapter 2) type them in the game chat with a leading `/`; the answer appears only in their own chat. Other players cannot run server commands and cannot change their own game mode. Pressing `/` in the game opens the chat with `/` already typed; `/help` lists the commands of the game client. While typing, the chat suggests commands, player names, item and mob ids: **Tab** completes, **Up/Down** picks a suggestion or browses earlier messages. In chat the player name is optional for `give`, `heal`, `health`, `power`, `air`, `gm` and `mob`: without it the command applies to yourself.

Command	Meaning
say <text>	Chat message from the server
players	List online players
kick <name>	Disconnect a player
ban / unban <name>	Ban (also offline players and their IP) or unban
mute / unmute <name>	Block or allow chat for a player
unregister <name>	Delete a player's password account (registration mode)
gamerule [name] [value]	Without arguments lists the rules: `pvp`, `explosions`, `explosion_blocks`, `explosion_players`, `explosion_claims`, `friendly_mobs`, `keep_inventory` (on|off), `death_drop_minutes` (1–1440), `max_ship_speed` (0–1000 m/s, 0 = GameSet). With a name shows or changes one (`gamerule keepinventory off`, `gamerule death_drop_minutes 5`); saved in `serverconfig.json`, announced in chat. Admins: `/gamerule pvp off`
gm <mode> <name>	Game mode: 0 survival, 1 creative, 2 spectator, or a mode of the GameSet by id or name (Default also has 3 `explorer`: survival without breaking or placing blocks). In chat an admin can leave out the name to change their own mode: `/gm 1`
tp <name> <x> <y> <z>	Teleport a player to a position: `~` is relative to the player (`tp Anna ~ ~10 ~`), `@a` means everyone. In chat the name is optional: `/tp 0 0 0`
tp <name> <target>	Teleport a player to another player (`teleport` and `tpp` work the same). In chat: `/tp Anna` teleports you to Anna
tphere <name>	In chat: teleport a player to you (`/tphere @a` brings everyone)
mob <id> <count> <name> | list | clear	Spawn, list or remove mobs
give <item> [count] <name>	Give items (ids like `iron_ingot`; `items [filter]` lists them)
setspawn [x y z] | setspawn clear	Set the spawn point of all players (new players start there, everyone respawns there); in chat without coordinates it uses the admin's position; `clear` goes back to a random asteroid for each new player. Players can go there with `/spawn`

protect / unprotect	In chat: protect the nearest asteroid or ship (up to 64 m from the admin) or remove the protection. On a protected one only admins can place and break blocks or fly it; nothing can damage or push it (bullets, explosions, collisions). Players get a red message. Good for a spawn area. `protected` lists them; saved in `protected_entities.json` in the world folder
heal <name>	Restore health, power and air
health / power / air [value] <name>	Restore one value to full, or set it
backup	Save the world and write a zip backup to `Worlds/_backups/` at once (also in single player from the console or chat)
stats	Server statistics: players online, most players today and ever (with date), joins, different players, time played by all players, server running time
save / stop / restart	Save the world; save and stop; save and load the world again (re-reads serverconfig.json, the console stays open)
clear, version, help	Clear the console, show the version, list commands

Claim beacons (player protection)

Players craft a **Claim Beacon** and place it on their ship (protects the whole ship) or their base on an asteroid (protects a cube of ± `Radius` blocks around it; the rest of the asteroid stays free to mine). They pay with ingots in the beacon's slot; every ingot adds protection time, up to the stored maximum. While time is left, only the owner, the members the owner adds in the beacon window and admins can build, break, open blocks or fly the ship there; doors still work for everyone, and bullets, bombs and collisions do no damage. When the time runs out, the area is free again, so abandoned claims disappear by themselves. Time only runs while the area is loaded on the server. Claims cannot overlap another player's active claim; `/protected` lists the loaded claims.

8. Files, saves and backups

Path	Content
Multiplayer/serverconfig.json	Server settings (also the server icon and this guide are in `Multiplayer/`)
Worlds/<WorldName>/	The world: sectors and asteroids, `world.nbt` (world settings), `Players/` (one `.player` file per player; all world files are NBT), `moderation.json` (bans, mutes), `accounts.json` (passwords in registration mode)
Worlds/_backups/	Zip backups made automatically before a world is converted to a new game version
Multiplayer/Dashboard/	`admin.json` (dashboard login), `setup-key.txt` (only before the first registration), `certificates/` (automatic HTTPS)
Mods/	`GameSets/` (content packs: blocks, items, recipes), `Extensions/` (extensions of a GameSet, chosen per world), `TexturePacks/` (texture packs; list ids in `TexturePacks` in serverconfig.json to send them to every player)

Backups
The server writes zip backups of the world to `Worlds/_backups/` by itself (see `Backups`), and `backup` (or `/backup` in chat for admins) writes one at once. To restore one, stop the server and unpack the zip in place of the world folder. Updating the game: copy the new game files over the old ones, keep `Multiplayer/`, `Worlds/` and `Mods/`. Older installs with `serverconfig.json`, `dashboard/` and `GameSets/` next to the executable are moved into the new folders on the first start. Worlds from 0.1.2 are converted automatically on the first start (a backup is made first); older worlds cannot be converted.

9. Troubleshooting

Problem	What to check
Friends cannot join	UDP 7777 forwarded to the right LAN IP; firewall rule; same game version; not behind CGNAT (chapter 4)
"Too many wrong passwords, try again in a few minutes"	5 wrong passwords for one name or from one IP lock logins for 5 minutes
A player cannot join: "The name ..."	Names may have at most 32 characters and no `\ / : * ? " < > |`, and cannot end with a dot
Dashboard does not open	TCP 7777 forwarded and allowed; you added `:7777` to the address; at home use the LAN IP
"Your connection is not private"	HTTPS is not ready yet or you opened `https://` with an IP: use the DuckDNS name; check the log for certificate errors
Certificate fails	The log line says why and what to check: DuckDNS/Cloudflare token, the name pointing to your public IP, TCP 80 forwarded (or `HttpsChallengePort`). `PublicAddress` must be public, not 192.168.x.x. It retries every hour; Let's Encrypt allows only a few failed attempts per hour, so test with `LetsEncryptStaging`
"Too many attempts"	5 wrong passwords: wait 5 minutes
Port already in use	Another program (or a second server) uses 7777: stop it or change `Port` / `DashboardPort`
World cannot be loaded	The log names the reason; worlds older than 0.1.2 are not supported